GEK-83864

MAINTENANCE / DIAGNOSTIC SUPPORT SYSTEM

(M D S S)

USERS GUIDE

DATE: May 19, 1989

AUTHORS: A B BLANKENSHIP
R E GILLMORE

Copyright (C) 1990 by General Electric Company.
Published only in a limited, copyright sense and all rights, including trade secret rights, are reserved.

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VIRGINIA 24153

MDSS Software Users Guide (GEK-83864)
Table of Contents

INTRODUCTION 1
Notational Conventions 1

SYSTEM OVERVIEW 2
Configuration 2
Design Criteria 3
Basic Operation 4
System Expandability 4

GETTING STARTED - BASIC MDSS CONCEPTS 5
How To Use The Computer Keyboard 5
Starting The MDSS Software 7
The Monitor Mode 8
Stopping The MDSS Software 9
The MDSS System Manager 10
Accessing The Other MDSS Functions 10
MDSS Menus And Prompts 12
How To Use The Menus 12
How To Use The Prompts 14
Priorities And Privileges 17
Exiting And Aborting MDSS Functions 18

MESSAGE LOGGING 19
Polling Of Control Products 19
Message Format 20
Types Of Messages 22
Product Event Messages 22
Application Messages 22
MDSS Internal Messages 23
Messages On The Video Terminal 23
The MDSS Printer 24
Secondary Printer Port 25
Local Printer Port 25

THE MDSS MAIN MENU 26
Using The Main Menu 26
Standard Main Menu Functions 26
Return To Monitor Mode (logoff) 27
Direct Access To Control Product 28
System Manager Functions 29
Set System Date And Time 29
Identify Current Software Module 30
Optional Main Menu Functions 31
Show Current MDSS Users 31
Alarm Annunciation Function 31
Review MDSS Message Log Files 32
Initiate Control Product Selftest 35
Message Explain Function 36
Change Explain Message Text 37
Change Application Message Text 40
Display Status Of Product Or System 43
Plot Control Product Circular List 44
Download A Control Product 47

ABOUT THE VAX COMPUTER 48
Computer Configuration 48
Loading The MDSS Software 49
Testing The MDSS Software After Installation . . 51
Backing Up The Vax System Disk 51

SYSTEM MANAGER INFORMATION 54
The System Manager User Account 54
Setting Up User Accounts And Priorities 55
Configuring An MDSS System 56
The System Manager Menu 56
Standard System Manager Menu Functions 57
Changing Username, Password, & Priority . . . 57
Modifying The System Configuration 59
Backing Up The MDSS Software 65
Stop Polling Of Control Products 66
Reset Configuration And Start Polling 67
Adding A New MDSS Software Module 67
Modifying Menu Text & Priority 68
Optional System Manager Menu Functions 69

INDEX

1 INTRODUCTION

This document is intended to familiarize the reader with the purpose and functions of the Maintenance and Diagnostic Support System (MDSS) software, and also serve as a reference document for more experienced MDSS users. Early sections of the users guide provide basic information concerning the operation and function of the system, while later sections give detailed operating instructions for the users and the MDSS System Manager.

The MDSS software requires that at least one person be responsible for managing the system. This person will be referred to as the "System Manager", and several sections of this document apply only to the System Manager. It is assumed that the typical user has no previous experience with computers, and that the MDSS System Manager has a basic understanding of the VAX/VMS operating system and VAX operating procedures.

The MDSS product is a software package which runs in a host computer. The combination of software and hardware together provide a powerful maintenance and diagnostic tool to its users. The MDSS software package is designed to run on a VAX computer under the VMS operating system (version 4.4 or later). Initial installation of the MDSS software will require someone familiar with VAX systems and VMS in particular. Once the software has been installed, anyone can use MDSS with minimum training; the software is menu-driven for simplicity, and basic instructions are always displayed on the terminal.

1.1 Notational Conventions

Some notational conventions are used throughout this document for improved readability. References to keys on the MDSS terminal will be enclosed by < > characters, for example, the <RETURN> key and the <TAB> key. Also, some control functions will be required; these are accessed by holding the <CTRL> key and then pressing the requested key. For example, <CTRL/A> means hold the <CTRL> key and then press the <A> key on the keyboard. Other conventions will be explained as they are used in this document.

On the newer VT200-Series terminals, the <DELETE> key has been replaced by a key with an "X" inside a box. This will be called the <DELETE> key throughout this document. The key is located above the <RETURN> key on the VT200 keyboard.

2 SYSTEM OVERVIEW

MDSS stands for "Maintenance and Diagnostic Support System". The basic purpose of the MDSS is to assist in reducing the down time of a process control system. This is accomplished by collecting events and faults which occur in the various components of the process control system and reporting them to one central location. These reports contain the information obtained within the controls of the system, as well as additional information detected within the MDSS host computer itself. They help diagnose problems and speed the return of the process to normal operation.

2.1 Configuration

One or more "control products" work together to form a "process control system". The MDSS has the capability of simultaneously performing its maintenance and diagnostic functions on multiple "control products" of multiple "process control systems", all from a central location.

"Control products" are devices which use micro-processors to perform fixed or programable functions within a "process control system". These control products can cover a wide range of applications, from digital regulation of a motor to control of a complex programmed sequence of events in the overall system. Each "control product" has a "port" which is used to communicate with the MDSS. A "process control system", then, is a collection of "control products" designed to perform a specific task.

As an example, a "process control system" might be a production line which manufactures widgets. This production line is controlled by a Series Six programmable controller and two digital drive regulators, for a total of three "control products". The function of the MDSS would be to monitor the programmable controller and drive regulators and report any problems detected within them. If the production line stopped making widgets, the MDSS should help the operator determine why the process stopped and what actions should be taken to get the process started again.

It is important to understand that the MDSS itself is NOT a "control product". It provides interaction with the control products in a manner which does not interfere with the process control system. The MDSS is, by definition, a maintenance and diagnostic tool, and therefore has no control over the process itself. In the above example, there would be a separate communications link between the programmable controller and the drive regulators to provide process control. The links between the MDSS and the control products carry only maintenance and diagnostic data.

A typical MDSS configuration consists of a host VAX computer, which is connected by a series of RS-232 cables to each control product port. This link to each control product from the MDSS replaces the terminal which would normally be connected to the control product; instead of having many terminals scattered throughout a plant, control products can be accessed and data gathered at one central location. All functions which can be performed by a local terminal connected directly to a control product are also available from the central MDSS terminal.

2.2 Design Criteria

The MDSS is designed for use by individuals with little or no computer experience. To simplify interaction with the user, the MDSS is a "menu-driven" system. This means that at any point, actions are selected from a "menu" of all possible actions available to the user. This minimizes the typing required to use the system, eliminates having to remember commands and keywords, and makes the system very easy to learn and understand.

The user is always presented with a menu of all functions he may perform at any given time. To perform a function the user simply selects the desired action from the menu. This is accomplished by using the arrow keys to position the blinking "cursor" to the desired item, then pressing the <RETURN> key. (Refer to the section on "How To Use the Menus" for more information). Protection is provided to insure any function selected will not interfere with any control system operation.

Occasionally, the MDSS will require information from the user which cannot be entered through a menu. In this case, a "prompt" is presented telling the user what information is required, and a space is provided for the user to enter a response. The MDSS carefully checks any data entered in this manner, and the user is provided a means of easily changing the response if needed. (Refer to the section on "How To Use the Prompts" for more detailed information).

2.3 Basic Operation

Each control product connected to the MDSS contains internal diagnostic information which monitors the status of its own hardware and software, if applicable. When a problem is detected, or some other significant "event" occurrs within the control product, the product sends a message to the MDSS. These unsolicited "event messages" are collected by the MDSS and reported to the user. It is through the reporting of these events and faults, and the user's knowledge of their meaning, that problems can be located and corrected in the shortest possible time, thereby reducing system down time.

Each control product sends its event messages to a "port" for output. The MDSS gathers these messages via a cable connection (link) to these ports. A "local terminal" may be connected directly to any control product port and used to read the messages locally at the control product, if desired, by disconnecting the MDSS link connection from that product. The "local terminal" may be any device that will display messages using an RS-232 type interface.

2.4 System Expandability

The system may be expanded to provide numerous other functions. New functions are designed as their needs are defined and the feasibility of their implementation studied. These new function may be added to an existing MDSS system as they become available. This capability of the MDSS to "grow" as new functions are developed is a very powerful feature; it will allow the system to become more advanced as new diagnostic techniques are developed, and provides a means of keeping the MDSS product up-to-date with the latest technology. Refer to the "Add A Module" section for more information on adding new modules.

3 GETTING STARTED - BASIC MDSS CONCEPTS

This section will help the user get started with the MDSS. It provides basic information on how to use the video terminal and keyboard, and general information on using the MDSS software. Details on selecting menu items and answering prompts are provided, which apply universally to all MDSS functions.

3.1 How To Use The Computer Keyboard

The operation of the computer keyboard has been standardized within the MDSS to insure it is easy to use and understand without any prior knowledge of computers. Simple operations and procedures have been established that apply to all functions of the MDSS. In an effort to minimize the training required to use the MDSS, basic functions can be accessed by simply following the instructions on the video terminal at each step.

The most basic concept of the MDSS is that all data entered must be terminated with the <RETURN> key. This key is used to transfer data to the MDSS after the user has entered it. In most cases no action will take place, and the data entered can be modified, until the time the <RETURN> key is pressed.

When entering data or selecting functions on the video terminal, a flashing block, called a "cursor", is used to show the current position on the display. The cursor always shows the position that is being "pointed to" by the MDSS. There are four "arrow" keys which can move the cursor around on the video terminal; <up arrow>, <down arrow>, <left arrow>, and <right arrow>. These keys will move the cursor in the direction indicated, if that is a legal operation at the time. If the cursor cannot be moved in the direction selected by the user, the terminal will "beep" and the cursor will not move.

A method has been provided to abort any operation by holding down the <CTRL> keyboard key and striking the <A> key. This action is called <CTRL/A> and will be referred to throughout this document as "Pressing the <CTRL/A> key". This will always return the MDSS to its previous function or operation.

There are only five basic keys that need to be learned to use the MDSS. These keys are described below:

1. <RETURN> key passes data to MDSS function requesting it OR selects item of a menu. ALL DATA ENTRY MUST BE TERMINATED BY THE <RETURN> KEY.

2. <LEFT ARROW> moves cursor left without changing any data.

3. <RIGHT ARROW> moves cursor right without changing any data.

4. <UP ARROW> moves cursor up one line should one exist.

5. <DOWN ARROW> moves cursor down one line should one exist.

Invalid entries will be signalled by a "beep" and will not be accepted. The user should enter valid data and continue.

The MDSS sometimes asks questions which require a YES or NO response. These questions always end with the text "(Y,N) ?", which means "please answer YES or NO". For example:

```
Do you want to stop polling (Y,N) ? _
```

The MDSS is designed so that only the <Y> key is required for a 'YES' response, or the <N> key for 'NO'. Note that upon pressing the <Y> or <N>, the MDSS completes the remainder of the word. Answers can be changed by just typing the desired <Y> or <N> over the previous entry. As always, the answer is not accepted into the MDSS function until the user depresses the <RETURN> key.

For additional information on how to use the menus and the prompting function, see the sections on "How To Use the Menus" and "How To Use the Prompts" in this document.

3.2 Starting The MDSS Software

The MDSS software package must be started once after the system has been booted. This is done by "logging in" to the VAX; press the <RETURN> key (usually once or twice) until the prompt "Username:" appears. At this point, type "MDSS" and press the <RETURN> key again. The MDSS software will now be started. The MDSS account should not require a password, unless the System Manager has assigned one.

Several messages will be displayed as the MDSS software is started. A normal startup after a VAX system boot will cause the following messages to be displayed:

```
%STARTMDSS-I-BEGIN, begin startup of MDSS tasks
%STARTMDSS-I-SMPT, starting Message Processing Task
%STARTMDSS-I-SLNK, starting the Link Task
%STARTMDSS-I-MDSS, starting the MDSS Monitor Task
```

After the sequence completes, the Monitor Mode is automatically entered. Several MDSS event messages will be produced with date/time stamp to indicate the MDSS has successfully initialized. These event messages are also logged to the printer.

This procedure for starting the MDSS software only needs to be performed after the VAX has been rebooted. Once the MDSS software has been started, it should never need to be restarted until the next system boot. If the MDSS software encounters an unrecoverable error during its initialization, an appropriate message will be displayed, and the System Manager should follow the instructions in the message.

The MDSS printer should be set to the top of a page before initiating this procedure. This is accomplished by pressing the LOCAL PAGE FEED button on the printer (or FORM FEED button on some printers), and then manually positioning the paper to print at the top of a page. Once this operation has been performed, the MDSS software will automatically page-feed the printer, as long as the user does not manually alter the paper position.

3.3 The Monitor Mode

The basic operating mode of the MDSS system is the "Monitor Mode". This is the mode that is initiated upon start-up of the system. It is also the mode the MDSS reverts back to when no other function is being performed by any user.

The Monitor Mode display is divided into three sections on the video terminal as follows:

1. the TOP of the screen displays the Monitor Mode header, copyright information, etc.

2. the MIDDLE of the screen displays fault and event messages (initially 5 lines).

3. the BOTTOM of the screen is reserved for the user interaction.

As the MDSS detects events within the control system, messages are logged on the printer with a time and date stamp. These messages are also displayed on the middle section of the video terminal when the MDSS is in Monitor Mode. This provides the user with a a fast reference to the printed log without having the look at the printer output. The terminal also provides an audible alarm (two "beeps") each time a new message is added to the display to notify the user. Note that for long messages, only the first line of the printed message will be displayed on the video terminal to conserve space.

If the alarm annunciation function is active, important messages will cause the printer to beep continuously at a preset interval. This is to notify the operator that an important alarm is present and has not been acknowledged. To acknowledge all outstanding alarms and stop the beeping, press the <PF4> key on the MDSS video terminal. This will also cause a message to be logged indicating the time and date that the alarms were acknowledged.

As events and faults occur, messages in the middle section will scroll up to make room for the new messages at the end of the displayed list. As these existing messages scroll up they push the copyright lines off of the top of the screen, expanding the message portion of the display to a maximum of 18 messages. Only the "Monitor Mode" title may be left in the top section.

Once a user has logged into the MDSS system the middle video terminal section will NOT be used to display event and fault messages. Instead the most recent message will be displayed on the bottom line of the video terminal regardless of any other functions being performed. The system is designed to insure all messages are displayed with as little delay as possible and no message is ever lost.

With a mulit-user MDSS system, it is possible to have one or more users in the Monitor Mode while other users may be logged into the MDSS under different usernames. All messages are displayed on all MDSS users' terminals as well as on the MDSS printer(s). Each user can be performing a different function; the only restriction is that two users cannot be accessing the same control product at the same time.

When the MDSS is in Monitor Mode, the terminal is not "logged in" to the MDSS. This is the "safe" mode, since no functions can be initiated from the Monitor Mode. In order to acess any of the other MDSS functions, a user must first "log in" to the MDSS by entering a valid username and optional password. This will be explained in a later section of this document.

3.4 Stopping The MDSS Software

Once the MDSS software has been started, it is never stopped under normal conditions. Users can leave the Monitor Mode by logging in to the MDSS and using the functions of the Main Menu. Upon logging off, the user returns to the Monitor Mode again. This is the normal sequence of operation for the MDSS software.

If the user wants to leave the monitor mode and disconnect from the VAX, the <CTRL/A> key will perform this function. This leaves the background MDSS tasks active, but terminates the Monitor Mode session. This would enable other MDSS users to start the Monitor Mode at different terminals by logging in using the MDSS username, as described earlier. This is the normal method of logging off of the VAX from an MDSS account.

If, for some reason, it should become necessary to completely stop the MDSS software package from executing, the user can do this from the Monitor Mode by holding the <CTRL> key down while typing the word "EXIT". This will

cause the polling sequence to be aborted, several messages will be logged to the printer, and the VAX will stop all of the MDSS processes from running. The terminal is now a regular VAX terminal, and the MDSS software in no longer active. To restart the MDSS software, follow the procedure outlined in the section "Starting the MDSS Software" of this users guide.

3.5 The MDSS System Manager

A user designated as the "System Manager" is allowed to assign levels of capabilities, called "priorities", to each MDSS user. These priorities control the functions each user may select from the menus. Those functions which a user does not have access to are automatically removed from the menus presented to the user.

The System Manager is also responsible for assigning usernames and passwords for the MDSS. Any problems encountered while using the MDSS software should be refered to the System Manager.

3.6 Accessing The Other MDSS Functions

The basic function of the MDSS is the Monitor Mode. Access to all other MDSS functions is permitted only through the Monitor Mode. The bottom line of the video terminal displays "Hit <RETURN> to access MDSS functions...". To access the functions the user must "log in" to the MDSS by performing the following actions:

1. Request access by pressing the <RETURN> key as indicated at the bottom of the video terminal. The prompt "Please enter your user name: ..." will then replace the previous message.

2. Type the assigned user name on the keyboard, and press the <RETURN> key. A "null" user name is permissible provided it has been assigned by the System Manager. To enter a "null" user name, simply hit the <RETURN> key in response to the prompt.

   If a user name is not entered within a preset time period, the MDSS will automatically revert back to the Monitor Mode function. An invalid entry will produce a message on the video terminal that "User (user name) is not known to the MDSS". This message will NOT be printed on the printer. The MDSS automatically

returns to the Monitor Mode after a timeout.

3. If no password has been assigned, proceed to the next step. Otherwise, the prompt "Please enter your password:" will replace the previous prompt. Enter the assigned password followed by the <RETURN> key. Note that the password is not displayed on the video terminal as it is entered; the <DELETE> key can be used to back up and correct mistakes, and <CTRL/U> (holding the <CTRL> key and pressing the <U> key) will delete all of the characters entered and start the password over again.

4. Correctly completing the above log-in sequence will display "Valid password..." briefly at the bottom of the video terminal until the MDSS is ready to display the Main Menu.

Anytime a user gains access to MDSS functions via this "log-in" procedure the following MDSS internal message will be printed on the printer:

(date/time) LI User (username) logged into the MDSS

It will also be displayed on the bottom line of the video terminal when the Main Menu is displayed.

As the user name and password are being entered corrections may be made as detailed in the next section "How To Use the Prompts".

3.7 MDSS Menus And Prompts

The user interacts with the MDSS through both menus and prompts. When the user must select from a list of possible choices, a menu is used. When the list of possible choices is too large to be presented as a menu, then the user will be prompted to type the required data on the video terminal keyboard. Menus and prompts always have the same format, no matter which MDSS function the user is performing at the time. The following sections describe how to use both the menus and the prompts.

3.7.1 How To Use The Menus

Menus provide a simple way for the user to select one of a group of choices. The "selecting" is performed by using the arrow keys to position the blinking cursor to the desired item and then pressing the <RETURN> key to select the item. The first time the user enters the Main Menu, these instructions will be displayed on the terminal as a reminder.

Menus will be presented as either a single or double column list, depending on the number of items in the menu. If the number of items in the menu exceeds one screen, then the menu will be automatically paged. This means that the first page of items (one screen full) is displayed, with the last item on the page being a link to the next page. If the desired item is not on the page shown, select the last item in the menu to display the next page of the menu. Both forward and backward menu links are provided in paged menus.

That is all the beginning MDSS user needs to know about using the menus. There are some additional selection features built into the menus, however, which the more advanced MDSS user can take advantage of. The following page contains a complete list of functions available to the user while in a menu.

Valid Keys For the Menu Functions

1. <RETURN> key selects item of the menu which the cursor is currently on.

2. <left arrow> moves the cursor to the left column item if one exists.

3. <right arrow> moves the cursor to the right column item if one exists.

4. <up arrow> moves the cursor up one item OR to the end of the menu if the cursor is currently on the first menu item.

5. <down arrow> moves cursor down one item OR to beginning of the menu if the cursor is currently on the last menu item.

6. <SPACE BAR> moves cursor down one item OR to beginning of the menu if the cursor is currently on the last menu item (same as <down arrow>).

7. <CTRL/A> aborts the current menu and returns to the previous menu.

8. <CTRL/R> re-displays the menu on the video terminal (this allows clearing up a display that may have been written by false data due to noise, etc).

9. all other keys - pressing any other key will position the cursor to the next menu item beginning with that character, if one exists. If no item exists with the specified key, the terminal will beep and the cursor remains on the previous menu entry.

10. <PF4> key can be used to acknowledge an audible alarm while in a menu.

NOTE

<CTRL/?> means hold the <CTRL> key down and then press the key specified by the "?".

3.7.2 How To Use The Prompts

An MDSS prompt always takes the form:

Prompt: _____

where the "Prompt" is a message telling the user what he is required to type, and the line is provided for the user to enter the response. An example of a prompt is the request for a username when the user presses the <RETURN> key in the Monitor Mode.

Please enter your user name: _____

The cursor will be placed at the beginning of the line, waiting for the user to type the requested data. As each key is pressed on the keyboard, the data is entered on the line, and the cursor advances to the next position. The length of the underlined field shows the maximum length of data that can be entered. If characters are entered after the field is full, they will not be accepted and the video terminal will beep to note the error.

If the user hits the wrong key, the <DELETE> key (the key above the <RETURN> key on the VT220 keyboard) will move the cursor to the left and remove the character from the line. The user can continue to enter and delete the characters on the line until the data is correct. At this point, the user should press the <RETURN> key to tell the MDSS that the data is correct.

That is all the beginning MDSS user needs to know about using the prompts. There are some additional editing features built into the prompts, however, which the more advanced MDSS user can take advantage of. The following page contains a complete list of functions available to the user while entering data at a prompt.

Valid Keys For the Prompting Function

1. <RETURN> key passes data to MDSS and terminates the prompt. DO NOT PRESS THE <RETURN> KEY UNTIL THE DATA ON THE PROMPT LINE IS CORRECT.

2. <DELETE> key deletes the character immediately to the left of blinking cursor.

3. <left arrow> moves the cursor to the left without changing any data.

4. <right arrow> moves the cursor to the right without changing any data.

5. <up arrow> moves the cursor up one line (if there is one).

6. <down arrow> moves the cursor down one line (if there is one).

7. <CTRL/A> abort the prompt and return the user to a previous operation.

8. <CTRL/B> moves the cursor to beginning of the field.

9. <CTRL/D> deletes all data from cursor position to the end of the field.

10. <CTRL/E> moves the cursor to the end of the field.

11. <CTRL/R> re-displays the prompt and data to the video terminal (this allows clearing up a display that may have been written by false data due to noise, etc).

12. <CTRL/U> deletes all of the data on the line and starts over. If data was initially present on the line, the initial data is restored.

13. <PF4> key can be used to acknowledge an audible alarm while answering a prompt without disturbing the data being entered.

NOTE

<CTRL/?> means hold the <CTRL> key down and then press the key specified by the "?".

When the user is prompted for data, text can be entered and modified as desired until the <RETURN> key is pressed. The cursor can be moved around within the underlined field using the four arrow keys, and text can be inserted at any point by typing the other keys on the keyboard.

When prompted for data entry or during editing of data, any other key depressed will insert that character in the position indicated by the cursor. If a character is already present under the cursor, that character and the cursor itself will move to the right one space, and any data following the cursor will also move right. Note that once the field is full, no more characters can be entered and the video terminal will beep until either the <RETURN> key is struck, or a character is deleted. The user can experiment with this "prompting function" by pressing the <RETURN> key in the Monitor Mode and using the "Enter user name..." prompt.

Some prompts are used for CHANGING data. When the user is presented with one of these prompts, the line will already be filled in with the OLD data. The user can modify the data presented using the keys previously described, delete the old data and enter new data, or simply press the <RETURN> key to leave the old data alone. (Since the cursor is still placed at the beginning of the field, a quick way to delete the old data is to use the <CTRL/D> function described above).

3.8 Priorities And Privileges

A security system based on passwords and priorities is incorporated within the MDSS to control which functions an MDSS user is permitted to use. Assignment of these priorities is usually performed by the MDSS System Manager.

Priority numbers in the range of 1 through 99 are used to determine access privileges. Priority numbers are assigned to each item in the Main Menu, and also to each MDSS user. A user can access all menu items which have a priority equal to or lower than the user's own priority. The System Manager may change the assignment of priorities for menu items and for users.

The higher the priority number of a user, the more functions that are available to the user. The higher the priority number of a menu item, the more protected that item is. Any menu items with a higher priority then the user's own priority will not appear in the menus for that user.

For example, a user with an assigned priority of 7 will be granted access to any menu item which has been assigned a priority between 1 and 7. Menu items with priorities above 7 would not be presented to this user in any menu, and therefore could not be accessed by this user.

Levels 1-98 can be assigned to users and menu items and have no special meaning. The System Manager can assign any priority in this range to any user or menu item as desired. Priority level 99 is the System Manager level. Since this is the highest priority that the System Manager can assign to any menu item, a user with this priority will be able to use any menu item. The System Manager can also assign a priority if 99 to other users, if desired; these users will then have the same access as the System Manager. Note that System Manager user can not be deleted, and the priority cannot be changed.

3.9 Exiting And Aborting MDSS Functions

A universal method has been provided to abort any action by holding down the <CTRL> key on the keyboard and pressing the <A> key. This will return the MDSS to its previous operation. At any point, if the user does not wish to continue the current operation, this method will safely abort the operation and return the user to some logical point, usually a menu. This combination (holding the <CTRL> key and pressing the <A> key) is called "Control/A", and will be shown in this document as <CTRL/A>.

Each menu also provides an easy method to return to the previous function, whether it be another menu or another mode of operation. Usually the first entry in the menu will be an exit function, which will return the user to the previous menu. An alternative method of exiting a menu is to use <CTRL/A> as described above. Pressing the <CTRL/A> key enough times will return the user to the Main Menu from anywhere within the MDSS software. Pressing <CTRL/A> while in the Main Menu will always return the user to the Monitor Mode.

4 MESSAGE LOGGING

The primary function of the MDSS is the reporting and logging of messages. These messages describe "events" which occur in the process control system. Some of these events may be problems, or "faults", within a control product, while others may simply be informational reports. These messages are gathered from all control products which are connected to the MDSS and reported to one central location for use by maintenance personnel.

All event and fault messages are logged on the MDSS printer to provide a permanent record, and also displayed to the video terminal for quick reference by the user. Both methods report messages from all control products that are functioning and connected to the MDSS. These messages are transmitted by the control product to the MDSS whenever the control product detects a reportable event. This function is always active, and messages are logged to the MDSS printer as soon as they are sent from the control product.

4.1 Polling Of Control Products

In addition to this message logging, the MDSS also "polls" each control product periodically to determine if it can still communicate with it. This "polling" function consists of the MDSS asking each control product to respond to a simple "polling request". The MDSS then waits for the control product to answer the request, and compares the response with an expected response. If the control product answers, then the MDSS continues polling other control products.

If a control product does not answer the request within a preset "polling delay", the MDSS logs a message saying that "control product (name) is not answering polling requests and was taken OFFLINE". This means that the MDSS cannot communicate with the specified control product any more. Polling will continue, however, and if the control product responds correctly at some later time, the MDSS will report that "control product (name) is responding to polling requests and is now ONLINE".

4.2 Message Format

As each message is received by the MDSS, the current date and time are added to the message. It is important to note that the date & time are added upon completion of receipt of the entire message from the control product. It therefore DOES NOT represent the exact time of occurrence of the event, but the time of receipt of the complete event message by the MDSS.

In addition to the date & time stamp, the MDSS adds to the message a "Product Identification" and a "Message Code", and re-structures all messages as necessary to fit on the MDSS printout. A complete message on the printer has the following format:

Date:	Time:	Prod:	Code:	Event Message:
30-JUL-85	09:49:12	MDREG	B0	BRIDGE CURRENT BAL FAULT
30-JUL-85	09:50:03	*MDSS*	OF	Product MDREG is OFFLINE
30-JUL-85	11:55:06	*MDSS*	LI	SYSTEM MANAGER logged in
30-JUL-85	13:10:23	MAINPC	E2	BASE SYSTEM RAM FAILURE ON THE HXYZ PROCESSOR

The various parts of the message are described as follows:

1. Date & Time Stamp - "Date:" and "Time:"

   Event and fault messages are date and time stamped upon receipt by the MDSS. The date & time are added upon completion of receipt of the entire message from the control product. Time is specified in 24-hour format (3:45 PM is logged as 15:45).

2. Product Identification - "Prod:"

   This field identifies the source of the message. If the message came from a control product, then this will be the control product identification which was assigned when the system was configured. This identification may be changed as desired by the System Manager (See the section on "Modifying the System Configuration" for details on how to change this). If the message was generated by the MDSS, then the message is identified by the "*MDSS*" in this field.

3. Message Code - "Code:"

   This 2-character code is used to classify the message. The code is assigned within the control product, except for those products that transmit "application messages". This allows duplication of codes between control products. The code is used by some of the MDSS functions which classify messages, and is also used by the "Message Explain" function.

4. Message Description - "Event Message:"

   This is the description of the event or fault that occurred. The message text is assigned within the control product and is printed exactly as it is received, with the exception of "application messages". If the message text is too long to fit on a single line, it will be continued on several lines (see the last example message above)

The basic structure of event and fault messages is the same for both the printer and the video terminal, except that messages displayed on the video terminal may be shortened to limit the display to a single line. Any portion of the message that exceeds one line is not displayed on the video terminal, but will still be printed on the printer.

Due to differences in the lengths of messages and the transfer rates from the control products it is possible to receive several messages from one control product during the time just one is being received from another. Although all control products are being monitored for messages, no action is taken until the end of a message is detected. Therefore, the ordering of the messages on the printout does not necessarily reflect the order of occurance of the events between different control products. Ordering of events within a single control product, however, is preserved.

This document does not describe any messages generated within the control products. Please refer to the documentation for each individual control product for a complete description of each control product message.

4.3 Types Of Messages

There are three types of messages which are reported by the MDSS:

1. Product Event Messages

2. Application Messages

3. MDSS Internal Messages

All three types of messages have the same format on the printed log, but the text of the message is created differently depending on the message type. To understand the terminology used in some of the MDSS function descriptions, it is useful to understand how the three message types are produced.

4.3.1 Product Event Messages

Product Messages are messages which are contained entirely within the control product. These messages are sent to the MDSS as a result of an event or fault in the control product, and detected by diagnostic software within the control product. The message is reformatted in the MDSS to produce the standard message structure detailed previously, but the text of the message is NOT altered.

The text and message code for Product Messages are stored in the individual control products and cannot be changed by the user.

4.3.2 Application Messages

Application Messages are optional specialized messages which originate ONLY within the application software of programmable control products. This application software depends upon the process control system software for event and fault detection. The control product application software produces an "Application Code" which is sent to a the MDSS. The MDSS uses this code to find the associated "message text" and "message code" in a table. This message text and code may be changed by a user privileged to make changes (see "Application Message Editor" section).

The text and message codes for Application Messages are stored in tables in the MDSS and can be changed using the Application Message Editor. Application messages are an option and may not be present in all MDSS systems.

4.3.3 MDSS Internal Messages

MDSS Internal Messages are generated when the MDSS detects an event without the help of a control product and generates a message. These messages are identified by a product identification of "*MDSS*". They usually describe events and faults within the MDSS, but sometimes the MDSS will detect unreported faults within control products and will report these as MDSS Internal Messages. For example, when the MDSS cannot communicate with a control product (the product does not answer an MDSS polling request) this is reported as an MDSS Internal Message.

The text and message codes for MDSS Internal Messages originate within the MDSS software and cannot be changed by the user.

4.4 Messages On The Video Terminal

In the Monitor Mode, the MDSS displays on the video terminal the latest 18 events & faults occurring in the control products. When the MDSS is being used for other functions the messages are stored for future display when the MDSS returns to the Monitor Mode. Although the video terminal may be in use, the most recent event/fault message is always displayed at the bottom of the video terminal. Hardcopy printing of all events & faults will occur independent of the mode of the video terminal.

When messages are displayed on the MDSS video terminal, two "beep" sounds are produced. This will notify the user to check the bottom of the video terminal for the new message. Also, the signal will provide an audible method of detecting when the printer should be operating; a line of printout should coincide with any double "beep". In the event that the printer were out of paper or otherwise disabled, the user could detect this.

As explained earlier, all messages are limited to a single line on the video display, even though the full message text is logged to the printer.

4.5 The MDSS Printer

A printer is provided to produce a hardcopy of all events and faults. A descriptive message will be printed upon receipt of an event or fault report from a control product, provided no other print action is in progress.

To help in detecting events and faults occuring in more than one control product, a line of dashes (----) will be printed to separate messages from different control products. For example, if several messages are received from product DMC001, and then an MDSS internal message is printed, the MDSS message would be separated from the DMC messages by a line of dashes. This makes reading the MDSS log easier, since messages often tend to be grouped according to products.

The MDSS software keeps track of the number of lines printed on each page, and will automatically advance the paper to a new page when it nears the bottom of a page. Each page will begin with an appropriate header to identify the page. For this reason, the user should NOT advance the printer paper by hand, since this will change the line number of the paper.

Event & fault messages are printed consecutively on a page. If a special function requires a printout, the printer will advance to the top of the next page and print an appropriate header. Any event or fault message occuring during printing initiated by this special function will be stored. The message will still be displayed immediately at the bottom of the video terminal, but message logging to the printer will be delayed until the special function has completed. An example of a special printer function would be printing the system configuration or listing the application message table to the printer.

Upon completion of the special function the printer advances to the top of a new page, prints the appropriate header, then prints all stored event/fault messages. It is recommended that any printing within a special function be implemented only when the control products are not operating, since this will delay the printing of fault messages.

If the alarm acknowledge function is installed and activated, important MDSS messages will cause the MDSS printer to beep continuously at a preset interval. This is to notify the operator that an important alarm is present and has not been acknowledged. To acknowledge all outstandind alarms and stop the beeping, press the <PF4> key on the MDSS video terminal. Refer to the section "Alarm Annunciation Function" for additional information.

4.6 Secondary Printer Port

A second MDSS printer port is also available. This allows the printed log to be available on two printers at two different locations, or provides a backup copy of the log at the central location, since the secondary printer port produces the same output as the primary printer port.

An alternative use of this secondary printer port is to provide the printed log data to another host computer for storage or analysis. The host computer must accept the data in the same format as it appears on the printer.

4.7 Local Printer Port

In addition to the normal MDSS printer, which records all event messages and faults, the MDSS video terminal may have a local printer connected as an option. This printer can be used to create a copy of any data currently displayed on the terminal by simply pressing the <PRINT SCREEN> key located on the keyboard. This key is also called the <F2> key (NOT the <PF2> key) on some terminals. The optional local printer must be present when the circular list plotting function is installed so that printed copies of the plots can be produced.

5 THE MDSS MAIN MENU

The Main Menu is the central core of the MDSS software. The Main Menu is what allows the user to select from all of the available MDSS functions he is authorized to use. Each item in the Main Menu is a separate function. Most of the functions are standard MDSS functions and will always be present in the menu, while other functions described in this document are options which may be purchased with the system or added at a later date.

Depending on how the System Manager assigns priorities to these menu items and to each user, all users may not have access to all standard MDSS functions. Therefore, all of the items listed in this section may not appear in the Main Menu for every user. See the section on "How to Use the Menus" for additional information.

5.1 Using The Main Menu

The items in the Main Menu select individual MDSS functions. These functions are described in the following sections. Note that the wording of the items in the Main Menu can be changed by the System Manager if desired, so the wording of menu items described in this document may not exactly match the menu for your system.

To select one of the functions from the Main Menu, follow the instructions printed on the video terminal, or read the previous section "How To Use The Menus" in this document.

5.2 Standard Main Menu Functions

The functions listed in this section are always part of an MDSS system. These functions represent the most basic functions that the MDSS can perform.

5.2.1 Return To Monitor Mode (logoff)

Selecting this function will return the user to the Monitor Mode. This function "logs the user off" of the MDSS by printing a message on the printer identifying the user. Once this function has been selected, the user will have to go through the "log-in" described previously (see the "Monitor Mode" section of this document).

This should always be the last function performed by all users before finishing with the MDSS video terminal. The user should NOT leave the MDSS in the Main Menu or any other function when they are finished using the system, since this allows other users access to the system without having to specify a password. Also, some MDSS Internal Messages which are printed contain the name of the user which is currently using the system. To maintain the integrity of the printed log, all users should return to the Monitor Mode (log off) when they are finished using the MDSS.

The "Return To Monitor Mode" function can also be activated from the Main Menu by pressing <CTRL/A>, the universal MDSS abort sequence. This will also log the user off of the MDSS and return to the Monitor Mode.

When the user returns to the Monitor Mode, all of the event messages which have been logged to the printer and displayed to the bottom of the video terminal during the MDSS session will be displayed once again to the terminal, beginning with the message which says the user logged into the MDSS, and ending with the logoff message. If more than 18 messages were printed, then only the last 18 will remain on the video terminal.

5.2.2 Direct Access To Control Product

Selecting this function will "connect" the MDSS video terminal directly to the specified control product. This allows the user to initiate any action which could be performed by disconnecting the MDSS link and connecting a terminal directly to the control product port.

Without the MDSS, the user would have to provide a video terminal (or printer) at every control product, or transport a terminal between control products. This terminal could then be attached to the port on the product and provide the user with the local diagnostic information for that control product. In addition, the user could converse with the control product through this "local terminal", asking for configuration information or setup parameters, for example, and possibly modifying these variables. This conversation would take place in a language unique to the individual control product.

With the MDSS connected to the product ports, there is no reason to have "local terminals" resident at each control product. Instead, this Direct Access function will "connect" the MDSS video terminal to the selected product automatically, through the MDSS link. Once this "connection" is made, the user can converse directly with the selected control product, in the language of that product.

This function gives the MDSS the ability to duplicate any control product action initiated from a "local terminal". This "direct access" feature has many advantages, some of which are: (a) the ability to communicate with any one of the MDSS control products from a central location, (b) not having to disconnect the MDSS communication cable from the control product, (c) not having to locate and transported a "local terminal" to the control product location, (d) centralized location for storing the documentation for all control products, and (e) the ability of selectivly printing the conversation between the control product and the MDSS.

When operating in the direct access mode the user must communicate in a language dictated by the control product. This language differs from one control product to another, and is the same language that must be used to communicate via a "local terminal" connected directly to the control product. Refer to the control product documentation for details of the control product languages.

To exit the Direct Access mode, enter <CTRL/A>. This will return the user to the Direct Acess product selection menu. At this time, another product can be selected for direct access, or the user can return to the Main Menu.

In a multi-user MDSS system, only one user may be in direct access with a given control product at a time. If a user attempts to access a control product which is currently being accessed by another user, an error message is displayed and the second user is denied access to the product.

To create a printed log of the session between the MDSS and the control product, enter <CTRL/P>. This will turn on the printer, and any data which is written to the video terminal will also be printed on the printer after the <CTRL/P> is entered. To stop the copying of data to the printer, press <CTRL/P> again. Any event messages which are received from other control products while the printer is being used by the direct access function will be displayed to the bottom of the video terminal immediately, but will not be logged to the printer until the user stops the print function with <CTRL/P> or exits the direct access function with <CTRL/A>.

5.2.3 System Manager Functions

This item takes the user to the System Manager Menu. For a complete description of the System Manager functions, see the section on the System Manager Menu.

5.2.4 Set System Date And Time

Selecting this function will allow the user to set the internal system time in the VAX computer. This is the time which appears on the printed log and the video terminal with every message. The VAX computer remembers the current time, even when the system is re-booted, but occasionally the user may want to modify the current time.

When this function is selected, the current time, rounded up to the next whole minute, is displayed on the terminal. The cursor appears as a large block positioned over the day-of-the-month part of the date. The user can change the day by pressing the <up arrow> key to increase the value, and the <down arrow> key to decrease the value. The other parts of the date and the time can also be modified in the same manner by pressing the <left arrow>

and <right arrow> keys to move the cursor to the desired part of the date.

Note that the time is entered in 24-hour format. The "AM" or "PM" at the end of the date cannot be changed by the user. It is updated by the MDSS system depending on whether the current time displayed on the terminal is before or after 12:00 noon. To set the time to 2:35 PM, the user must set the hour field to 14 (24-hour clock).

In addition to the increase and decrease functions, the user may also directly enter a value into the field. With the cursor positioned over the hour, for example, the user can type "12" on the keyboard, then press the left or right arrow key, and the hour value will change to 12. The month name can also be changed in the same manner, but the number of the month, rather than the name, must be entered (JAN = 1, FEB = 2, etc.)

When the time on the display is correct and the user is ready to change the system time to the time on the display, the <RETURN> key should be pressed. The function will then ask the user if the time shown should be saved as the system time. Answer <Y> to actually set the system time, or <N> to continue modifying the time on the display. The seconds are set to zero when the system time is set. If the user wants to exit the time set function without changing the system time, the <CTRL/A> key will return the user to the Main Menu without affecting the system time.

The <CTRL/V> key can be pressed at any time while setting the time and date to display the current software module name and version number.

5.2.5 Identify Current Software Module

This item only appears in the Main Menu for the System Manager (or other users with priority 99). It allows the System Manager to obtain the current MDSS software module name and its version number. This function is only used to report software problems to GE; it will appear as the last item in many of the menus for the System Manager user.

5.3 Optional Main Menu Functions

The functions listed in this section are options which can be purchased for an MDSS system. These functions add additional capabilities to the MDSS, but are not required for the MDSS to function. Depending on which functions have been purchased and how the System Manager assigns priorities to these menu items and to each user, all users may not have access to all optional MDSS functions. Therefore, all of the items listed in this section may not appear in the Main Menu for every user.

5.3.1 Show Current MDSS Users

The basic MDSS allows only one user logged into the MDSS account at a time. An option is available to support multiple MDSS users on the same VAX computer. If the multi-user MDSS option is present, this menu item will display all of the currently active MDSS users and show what they are doing. For example, the function may show two users are in Monitor Mode, one user is in direct access with control product XYZZY, and another user is running the MDSS Main Menu module (which implements this "Show Users" function).

With a mulit-user MDSS system, it is possible to have one or more users in the Monitor Mode while other users may be logged into the MDSS under different usernames. All messages are displayed on all MDSS users' terminals as well as on the MDSS printer(s). Each user can be performing a different function; the only restriction is that two users cannot be accessing the same control product at the same time.

5.3.2 Alarm Annunciation Function

A function is available to annunciate important alarm messages. This function will cause the MDSS printer(s) to provide an audible alarm when important event messages are received until the operator acknowledges them.

If the alarm annunciation function is active, important messages will cause the MDSS printer(s) to beep continuously at a preset interval. This is to notify the operator that an important alarm is present and has not been acknowledged. To acknowledge all outstandind alarms and stop the beeping, press the <PF4> key on the MDSS video terminal. This will also cause a message to be logged indicating the time and date that the alarms were

acknowledged.

The <PF4> key may be used to acknowledge alarms from any mode of MDSS operation. If a user has gained access to the MDSS by logging in, then the message will contain the name of the user who acknowledged the alarms. When the alarms are acknowledged from Monitor Mode, a general message is printed. In either case, the audible printer alarm is stopped.

This menu item allows the user to enable and disable the annunciation function, and define the delay between audible alarms. When this function is selected, the user is asked if the function is to be disabled. Answer YES to disable all audible alarms, or NO to enable the alarming. If the user answers YES, the MDSS will prompt for the delay time (in seconds) between alarms. Any number between 0 and 99 is valid, although a delay of 0 seconds is equivalent to disabling the function. Typical values for the delay are 1 to 5 seconds.

5.3.3 Review MDSS Message Log Files

This module will allow the user to sort and display message log files. The message log files contain all the event messages which have been displayed on the MDSS printer. The user can review these log files, either partially or totally, based on several parameters. The user can display all messages, just messages from a particular control product, or just messages with a particular message code. In all cases, the user can select a starting and ending date to bracket the messages which are displayed.

The main menu of the log file review function provides the following functions:

1. Exit - return to the previous menu

   This function exits the log file review function and returns the user to the MDSS Main Menu.

2. Select a new log date or date range

   This function allows the user to select the date of the messages to be reviewed. If only one log file exists, this option will automatically select that

file.

3. Display all messages in the date range

   This function displays all of the event messages which occured on the selected date(s) after the current starting time.

4. Display messages by product name

   This function displays the event messages which occured on the selected date(s) after the current starting time and were generated by the selected control product.

5. Display messages by message code

   This function displays all the occurences of a selected event from the specified date(s) after the current starting time. You will be prompted for a two-character message code to search for. One of the two characters may be a wildcard "*" character, which matches any character. For example, you can search for all messages with a message code ending with "X" by searching for the code "*X". You must also select the control product to search for; one of the items in the control product selection menu will be "All products", if you want to find all occurrences of a particular message across all control products...

6. Select a new starting time

   This function allows you to select a starting time (hours, minutes and seconds) for the current day. If you do not select a starting time, it defaults to midnight. Once you have selected a starting time, it is used to qualify all of the other selections. The current starting time is always displayed below the selection menu.

7. Print output from the last command

   This function causes the output from the last display command you issued to be copied to a local printer attached to your terminal (NOT to the MDSS printer).

While you are displaying a list of messages which match your selection criterion, you may abort the list at any time by pressing the <CTRL/A> key sequence. If you allow the listing to complete, the function will display a total count of how many messages were displayed.

The logical name MDSS_$LOGPURGE defines how many days worth of log files are saved. The user may set this logical name (defined in file LOGICALS.DAT in MDSS$_Gbldir:) to an integer number between 1 and 45, which corresponds to the number of days worth of log files to retain. Files which are older than this limit will be automatically purged. If the logical is not defined or is invalid, the default is 8 days of log files.

5.3.4 Initiate Control Product Selftest

This function will attempt to initiate a hardware selftest on a control product. The user will be asked to select which control product should be tested. The MDSS will then try to initiate the internal control product self-test algorithm. Each control product has the authority to accept or reject the self-test command from the MDSS. In most cases, if the control product is actively controlling the process, the self-test command will be rejected. The user should only attempt to initiate a self-test on control products which are not currently working.

When the user selects this function, a menu of all control products which have an internal self-test function is displayed. The user may select one of the products from this menu, or return to the Main Menu. If a control product is selected, the MDSS will notify that control product to initiate a selftest. If the product rejects the request, a message will be reported to the user. Otherwise, if the product accepts the self-test request, a message will be printed showing who initiated the selftest, the time and date of the request, and the control product name. At this point, the control product will go offline and begin its internal test.

At the completion of the self test sequence, the control product will report any errors detected to the MDSS. These will be printed and logged to the terminal. If no errors are detected, the control product will report that it is healthy, and the MDSS will (after a time delay) detect that it is ONLINE again.

This internal control product self-test is the most complete method available to the MDSS for testing the hardware and software of each control product. The self-test algorithms in the individual control products attempt to find a variety of internal problems, such as card failures, software checksum errors, communications errors, and power supply problems. Each control product type has its own unique self-test (or "powerup") sequence; refer to the control product documentation for additional details.

5.3.5 Message Explain Function

The Message Explain function provides the user with more detailed information about the messages on the MDSS printed log. This detailed information can be expanded by the user to include very specific comments, a history of previous occurrences of each fault, or any other text the user wishes to add (see "Change Explain Message Text" for information on how to do this). This function is provided as part of the Message Explain option.

When the user selects this function, a menu of all control products known to the MDSS is displayed. Also, an entry "*MDSS*" is provided for the MDSS itself. The user should select the control product from this menu which corresponds to the message(s) to be explained. After a control product is selected, the MDSS will prompt for the two-character code of the message to be explained. The MDSS will then display all of the information on that message that can be found. (See section on "Message Format" for a description of the Message Code).

If more than one screen of information is stored for the selected message, the user will be prompted to "Press the <RETURN> key to go to next page...". The MDSS can store up to two screens, or "pages", of text for each message. The MDSS internal messages can also be explained by selecting the "*MDSS*" product from the menu.

If the user selects a control product and message combination for which there is no information, an appropriate error message will be displayed and the user will be prompted to select another control product or message code.

Pressing the <RETURN> key in response to the prompt for a message code will exit the Message Explain function and return the user to the Main Menu. Pressing <CTRL/A> will return the user to the product selection menu.

5.3.6 Change Explain Message Text

Selecting this function will invoke the Message Explain Editor, which is used to create and modify explain messages. This function is provided with the MDSS system as part of the Message Explain option.

Explain messages are stored in the MDSS in two parts, called "Page 1" and "Page 2". When messages are displayed (using the Message Explain Function described above) the Page 1 text appears as the first screen, and the Page 2 text, if any, will be the second screen.

Page 1 messages are initially provided as part of the MDSS Message Explain option, and can only be modified by a user with System Manager priority (priority = 99). There is only one page 1 message table per product type; for example, all DMC's have the same page 1 message table. This is because all products of the same product type produce the same messages with identical message codes. Therefore, changing a page 1 message for ANY product of a given type will change that message for ALL products of that type.

The Page 2 messages are stored individually for each control product. This allows individual control products to have unique messages for products of the same type. For example, the user may include in the page two message table a description of where the individual control product is physically located within the plant. The System Manager can control who can modify the page 2 messages by assigning a suitable priority to this function. Any user who can access this function can change any page 2 message. Only users with a priority of 99 can modify the page 1 messages.

When this function is selected from the Main Menu, a user with a privilege of 99 will be asked "Do you want to edit the Page 1 explain table (Y,N) ?"; answer <Y> to edit page 1, otherwise answer <N>. All MDSS users with a lower priority will automatically be editing the page 2 table.

Once the correct page table is selected, a menu of all control products known to the MDSS is displayed. Also, an entry "*MDSS*" is provided for the MDSS itself. The user should select the control product from this menu for which messages are to be edited. After a product has been selected, the Explain Editor Menu is presented.

The Message Explain editor menu provides the following functions:

1. Exit - return to the Main Menu

   This function exits the Message Explain Editor and returns the user to the MDSS Main Menu.

2. Add explain message

   This function allows the user to add new explain messages to the table.

3. Change explain message

   This function will allow the user to change a message which is already in the table.

4. Delete explain message

   This function allows the user to delete a message from the table.

5. Display explain message

   This function allows the user to display a particular message or all of the messages in the table for the selected control product.

6. Print explain message

   This function allows the user to print a particular message or all of the messages in the table for the selected control product.

7. Select new control product

   This function allows the user to select a new control product table to modify.

Most of these functions are self-explanitory. When new message text is being entered or old text is being changed, the editor uses the standard MDSS "prompting" function to enter one line of the message at a time. The user can press the <up arrow> and <down arrow> keys to move the cursor from line to line. Message text does not automatically wrap from one line to the next; the user must format each line exactly as it is to appear on the terminal when the message is displayed. To save the message from the screen to the message table, press the <RETURN> key. See the section on "How To Use the Prompts" for additional information on using the prompting function of the editor.

5.3.7 Change Application Message Text

Selecting this function will invoke the Application Message Editor, which is used to create and modify application messages. This function is provided with the MDSS system as part of the Application Message option. For a description of how application messages are produced, see the section on "Application Messages".

This editor allows the user to create and modify the application message tables. Each control product which supports application messages has its own table of messages. There are three things which are stored in the table for each message; a "message key code", a "message code", and "message text". These message parts will be explained in greater detail below.

When the user selects this function, a menu of all control products which support Application Messages is displayed. The user should select the control product from this menu which corresponds to the application message(s) to be modified or added.

Once the correct control product table is selected, the Application Message Editor Menu is presented. The editor provides the following functions:

1. Exit - return to the Main Menu

   This function exits the Application Message Editor and returns the user to the MDSS Main Menu.

2. Add application message

   This function allows the user to add new application messages to the table.

3. Change application message

   This function will allow the user to change a message which is already in the table.

4. Delete application message

   This function allows the user to delete a message from the table.

5. Display application message

   This function allows the user to display a particular message or all of the messages in the table for the selected control product.

6. Print application message

   This function allows the user to print a particular message or all of the messages in the table for the selected control product.

7. Select new control product

   This function allows the user to select a new control product table to modify.

The "message key code" is the 4-character code which is produced by the control product when an event occurs. For example, suppose when a hydraulic pump fails, the logic in the DMC will send a key code of "0023". This is the key that the MDSS uses to find the text associated with the event. There can only be one message with a given key for each control product. The MDSS user never sees the message key code; only the programmer who creates the message detection logic in the control product, and the MDSS System Manager who uses this function need to be concerned with the message key code.

The "message code" is the usual MDSS message code which will appear on the printed log and on the video terminal. It is used by the Message Explain function to explain the message. The message codes are usually unique for each message, but they do not have to be. It may be desirable, for example, to have several similar application messages which all point to the same explain message; these messages should all have the same message code.

The "message text" is the actual text of the message which will be printed on the printer. This text may be up to 240 characters long for application messages. Using the example above where the DMC detects the hydraulic pump failure and sends a message key code of "0023", if the user wanted a message to be printed of the form:

(date & time) F7 HYDRAULIC PUMP #2 FAILURE

The message would be entered with a message key code of "0023", a message code of "F7", and the text would be "HYDRAULIC PUMP #2 FAILURE". The data is entered using the "prompting" function to enter each part of the message. The user can press the <up arrow> and <down arrow> keys to move the cursor from one message part to another, until the message is saved in the table. To save the message from the screen to the message table, press the <RETURN> key. See the section on "How To Use the Prompts" for additional information on using the prompting function of the editor.

5.3.8 Display Status Of Product Or System

This function will display the current status of a control product. The user will be asked to select a particular control product, or the entire MDSS system. The MDSS will then send a status request to the selected control product (or products) and display the results to the user. The purpose of this function is to find system faults which have not been reset within particular control products.

When the user selects this function, a menu of all control products which have an internal display status function is displayed. The user may select one of the products from this menu, the entire MDSS network, or return to the Main Menu. If a single control product is selected, the MDSS will send a display status request to that control product. If the entire MDSS network is selected, each control product will be tested individually. In either case, messages will be displayed to the terminal as each control product is tested.

Several possibilities exist when testing the status of a control product. If the control product is currently not responding to polling (OFFLINE), then the MDSS reports this fact and does not attempt to issue a status request to that product. Otherwise, the MDSS sends a status request to the control product and waits for a reply. A control product may respond with a sequence indicating that no outstanding faults were detected. A single line is displayed for each control product which responds in this manner. Other control products may report faults which require a reset or other operator action. These messages are displayed directly to the terminal to alert the operator.

This function does not test a control product as completely as the self-test function. The display status function only gives an overview of a control product or the entire MDSS system, noting whether each control product is offline or online, and whether or not it has outstanding faults which must be cleared. This function will usually not perform detailed diagnostics on control products.

5.3.9 Plot Control Product Circular List

This function will plot data which has been collected in a control product circular list. The Direct Access function or some other external means must be used to set up the circular list before this function can plot any data. Refer to the control product documentation for additional information on setting up the circular list. This function also requires a graphics terminal such as a VT240 to produce the data plots; if you invoke this command from a non-graphics terminal, the only function your will have access to is the "Collect new data" function...

When the user selects this function, a menu of all control products which have an internal circular list function is displayed. The user may select any product from this menu and plot data from this product, or collect new data from the control product.

After selecting a control product, the function will read parameter data from a special parameter file. This file defines the scaling and descriptive information to be displayed for each parameter which the user may wish to collect. Next, the function tries to read a file which contains the data from the last dump of the circular list, if one exists.

Finally, the main plotting menu is displayed. The user can choose from the following functions:

1. Exit - select a new control product

   This function exits the plotting menu and returns the user to the product selection menu.

2. Collect new data from control product

   This item will cause the MDSS to collect the current circular list data from the selected control product. The control product must currently be online, and a circular list must have been previously set up before this function can collect any data.

3. Reset all selected variables

   This item will reset all selected variables in the plotting menu. Any items in the menu which are

followed by a "Yes" will be changed to "No".

4. Plot the selected variables now

   This item should be selected to actually create the circular list plot of the current data. The user must have selected at least one variable from the menu, and no more than 4.

5. Select variable (variable name) ? Yes/No

   The rest of the menu will contain items which list the names of the variables which were collected in the circular list. If one of these items is selected, the "No" will become a "Yes", which means that this variable will be included in the graph. If a "Yes" item is selected, it will be reset to a "No".

For users with System Manager priority, the third menu item "Reset all selected variables" will initially be replaced by the "Identify current module" item. After the first plot has been created, this menu item will change to the "Reset..." item shown above.

Once the user has collected the data and chosen one or more variables from the menu, a plot can be created of the selected variables. If a printer is connected to the MDSS terminal, the user can copy this plot to the printer by pressing the <PRINT SCREEN> key on the terminal while the plot is displayed. To exit the plot and return to the plotting menu, press any key while the plot is displayed. Refer to the section "Optional Local Printer Port" for additional information on the local printer and <PRINT SCREEN> key.

Note that when circular list data is collected from the control product, it is saved in a file for that product. Once this data has been collected, it can be plotted and replotted as often as desired until data is again collected from the control product. The latest data from each control product is always retained in a file in the MDSS.

The circular list data is typically plotted against a scale from 0 to 100, where 100 represents 1-per-unit (100%) value. The user must interpret this data according to the values used to tune up the drive. For example, if Siltron armature current feedback (IAFB) has been recorded in the circular list, and a plot of this parameter shows values in the range of 80 to 100, this would correspond to 80% - 100% of the full armature current for this drive. The scale is automatically adjusted to compensate for values which exceed 100%, or for parameters which have only a limited range.

The circular list data is plotted against a common time base. There are two possible sample times in the drive which can be used to record the circular list data; slow and fast. Once the data has been recorded, the MDSS cannot determine which time base was used. The data will always be plotted against the slow (normal) time base, in units of seconds. If the data was recorded at the fast sample rate, the user must divide the time base by 6 to get the true sample time in seconds.

Messages which are normally displayed to the last line of the video terminal will be suppressed while the plot is displayed. This is to allow the user to create a printed copy of the plot without having the current message at the bottom of the plot. These messages are still logged to the printer as they occur, and will be displayed to the terminal when the user returns to Monitor Mode.

During the time a plot is being created, the video terminal keyboard will be disabled. Once the user has pressed the <PRINT SCREEN> key, the cursor will disappear from the display and the printer will begin creating the plot. A complicated plot may take several minutes to complete. Once the plot has completed, the cursor will return to the bottom of the display and the user may press the <RETURN> key to return to the plotting menu. There is no easy way to abort a plot once the <PRINT SCREEN> key has been pressed.

5.3.10 Download A Control Product

This function allows the user to download control product software through the same RS-232 connection which the MDSS uses for communication. This function currently supports downloading DMC control products only. The actual download operation takes place asynchronously in a background subprocess which is created by this function.

When you select this function from the Main Menu, you are presented with a list of control products which currently have download files defined for them. You must select the product you wish to download; this causes a subprocess to be created which will perform the download. The user is notified by the subprocess when the download has been initiated.

NOTE: For a DMC, you must manually take the DMC offline before initiating the download (using the direct access function)! If you do not do this, your download will fail with a "DMC is online" error.

Once the download has completed, a message is logged to the printer which shows the completion status of the download. The DMC automatically reboots itself with the newly-downloaded system (assuming it was successful) and you will see the startup messages from the DMC system on the printer.

If the download fails for any reason, you should receive a message explaining the nature of the failure. Additional information is also provided in the MDSS log directory in a file called DNL_product_name.LOG where "product_name" is the MDSS control product name. You should read this file (using any standard VAX editor) if you have recurring download problems for some control product.

Due to the background nature of the downloads, you may have several different control products being downloaded simultaneously. You will be prevented from downloading the same control product twice at the same time. You should NOT log out of the MDSS Monitor Mode while any downloads are active, because this will delete the subprocesses. If you try to do this, you will receive a warning message and be returned to the MDSS Monitor Mode.

6 ABOUT THE VAX COMPUTER

The MDSS software package runs on the VAX family of computers, which are manufactured by Digital Equipment Corporation (DEC) and have been in use for a number of years. The software runs under the VMS operating system (version 4.4 or later), although the MDSS user never interacts directly with VMS. The VAX family contains small systems, such as the Micro/VAX, on up to large mainframes such as the VAX-8800. MDSS software can use any of these systems, depending on the size of the process control system configuration, and the number of other functions being performed by the computer.

6.1 Computer Configuration

The MDSS software requires the following minimum VAX configuration:

* Any one of the VAX family of computers.

* VMS Operating System (version 4.4 or later).

* One VT-200 or VT-300 series terminal to serve as the MDSS operating terminal.

* One printer (LA50, LA100, etc.) dedicated to MDSS output - this cannot be the VAX console device.

* A minimum of 4 Megabytes of main memory.

* A minimum of 15K blocks of free disk storage for MDSS code & data.

* A TU81, TK70 or other tape media distribution device.

* One RS-232 port (or terminal server port) for each control product link.

This document will not attempt to cover the installation, testing, and startup of the VAX computer or the VMS operating system - please consult the appropriate DEC manuals for additional information on the VAX computer.

6.2 Loading The MDSS Software

The MDSS software is distributed on magnetic tape or other suitable distribution media, and must be loaded by the VAX manager. This section will provide the necessary information for a VAX manager to load the initial MDSS distribution.

The following procedure should be followed in loading the MDSS software from the distribution media:

1. Log into the VAX SYSTEM MANAGER account (or another account with full privileges).

2. Define logical names for the disk and distribution device as follows (substitute the device names for your system where appropriate):

```
$  Define Disk ????:  (use your disk name)
$  Define Tape ????:  (use distribution device name)
```

3. Run the AUTHORIZE utility and create an MDSS username with the following sequence (you may use a different UIC value if [301,001] is already assigned):

```
UAF> ADD MDSS_UIC /Identifier /Value=UIC:[301,001]

UAF> ADD MDSS /NoPassword /Prior=5 /NoPwdExp
/Owner=MDSS /Account=MDSS /UIC=MDSS_UIC
/Device=MDSS$DISK /Lgicmd=MDSSLOGIN.COM
/Directory=[MDSS] /Flag=CAPTIVE /Priv=Tmpmbx
/Defpriv=Tmpmbx /Bytlm=65536 /Diolm=255 /Biolm=255
/Enqlm=255 /Astlm=255 /Fillm=64 /Prclm=8
```

4. Create a directory for the MDSS software using the following command:

```
$  CREATE/Directory   Disk:[MDSS]   /Owner=[MDSS_UIC]
   /Version=0 /Protection=(S:E,O:RWE,G:E,W:E)
```

5. Load the distribution media on the appropriate device.

6. Mount the device /Foreign (volumn label should be MDSS).

7. Copy the files from the distribution device to the disk using the following BACKUP command:

```
$ BACKUP /Verify
_From:     Tape:MDSS.BCK /Save /Select=[MDSS...]*.*
_To:       Disk:[*...]*.* /Owner=MDSS_UIC
```

8. Dismount and unload the distribution device.

9. Edit the file Sys$Manager:SYSTARTUP.COM and add the following lines, substituting your disk name for ????:

```
$  Define /System /Executive MDSS$Disk ????:
$  @MDSS$Disk:[MDSS.Sysdir]SETUPMDSS.COM
```

10. You will also have to change some of the SYSGEN parameters. The suggested method of doing this is to add the following lines to the bottom of file Sys$System:MODPARAMS.DAT

```
TTY_TYPAHDSZ=256
TTY_ALTYPAHD=512
TTY_ALTALARM=384
```

After making these changes you should run AUTOGEN to automatically adjust other system parameters before rebooting the system. Refer to the VMS System Management Guide for further details about the VMS AUTOGEN procedure.

The MDSS software is now resident on your disk. You should reboot the computer system and check the console listing for MDSS initialization messages. (If you chose to execute DETACHSET.COM instead of SETUPMDSS.COM, your initialization messages may appear on the MDSS terminal instead of the console listing). The MDSS software will not be completely installed and ready to run until after

the VAX computer has been rebooted.

6.3 Testing The MDSS Software After Installation

After the VAX Manager has loaded the MDSS software as described above, the MDSS System Manager should test the software. This procedure also will allow the System Manager to assign a password to the MDSS System Manager account, which is distributed without any password.

Log into the VAX using the username "MDSS". No password should be required. The MDSS startup sequence described earlier should execute, and you should end up in the Monitor Mode. Press the <RETURN> key and enter a user name of "SYSTEM MANAGER". Press the <RETURN> key again when asked for the password, and you should be in the "Main Menu". Select the "System Manager functions" item in the menu, which takes you to the System Manager Menu. Select the "Change usernames and passwords" item. See the description of this function for additional details on how to change the password for the System Manager user, and also how to add additional users to your system.

If your system was pre-configured by GE before shipping, then you have finished installing the software. If you must configure the system, then see the section on how to use the Configuration Editor. You will also have to edit the file MDSS$_Gbldir:LOGICALS.DAT if you must configure the system, and add logical names of the form "MDSS_PORT_nn" to point to the control product ports. The "nn" part of the logical name is called the "port id" and will be used later by the Configuration Editor program. Also define your printer port and tape drive, as shown in this file.

6.4 Backing Up The Vax System Disk

One of the most important functions of any computer system is the "backup" function. Backups are required in case of failure of the system disk, or corruption of some of the data on the disk. While these occurrances are rare, they do happen from time to time and, without proper backup procedures, they can cause weeks or months of work to be lost.

Your MDSS system is distributed on magnetic media. If the VAX was purchased with the MDSS software, then the distribution media will contain a complete backup of the system disk, including the VMS operating system files and all MDSS software. If the MDSS software was purchased as an addition to an existing VAX, then the distribution media contains only the MDSS software. In either case, the MDSS data tables will certainly be modified as soon as the software is installed, as user accounts are created, the system configuration is modified, etc. Without proper backup procedures, the only solution to a disk failure would be to reload the original distribution media and start over.

IT IS STRONGLY RECOMMENDED THAT A PROCEDURE BE ESTABLISHED THAT DICTATES WHEN BACKUPS ARE TO BE MADE. THE FOLLOWING SHOULD BE CONSIDERED IN DETERMINING WHEN TO MAKE MDSS BACKUPS:

Backup the message tables & data files any time that:

1. the system configuration has been modified
2. any menu item text or priority has been modified
3. any user account has been modified (user added or deleted, password changed, or priority changed)
4. an application message has been modified
5. an explain message has been modified

Backup the entire MDSS system disk any time that:

1. a new module has been added
2. any operating system parameters have been modified
3. any other non-MDSS activity has created new files on the disk, or modified any files.

Note that it is NOT necessary to backup the disk unless one of the above activities has caused files to be modified. If the VAX is being used exclusively as an MDSS, then only after the user initiates one of the functions listed above will the disk need to be backed up. If you have other non-MDSS files on the MDSS disk, then

you will have to develop your own backup schedule.

It is also possible to perform disk backups using the VMS BACKUP utility. This method of performing backups is not covered in this document; refer to the VAX/VMS Utilities Reference Manual for additional information.

7 SYSTEM MANAGER INFORMATION

The MDSS software is controlled by a user designated as the MDSS "System Manager". This individual should have a working knowledge of VAX the computer and the VMS operating system, since he will be responsible for installing and maintaining the MDSS software. The remainder of this document is designed to provide the additional information required for the System Manager to perform these functions.

There are two main functions which the System Manager will usually perform; assignment of passwords and priorities to the users and menu items, and initial configuration of the MDSS. While both of these are relatively simple tasks to perform, they require a greater understanding of the MDSS than the average user needs to be concerned with.

7.1 The System Manager User Account

There is a user account designated as "System Manager" that cannot be changed except for the associated password. The System Manager has a fixed priority of 99. This allows the System Manager user access to all MDSS functions. Without this fixed account, it would be possible to change priorities of the menu items and users in such a manner that some functions would not be accessable to anyone and could never be recovered.

The system is shipped with a System Manager user name already assigned. It is accessed by entering a user name of "SYSTEM MANAGER". No password is required initially for this user (the MDSS will prompt for a password; simply hit the <RETURN> key to continue). IT IS STRONGLY SUGGESTED THAT THE SYSTEM MANAGER LOG INTO THE MDSS SYSTEM AND CHANGE THIS 'NULL' PASSWORD TO SOMETHING ELSE WHICH CAN EASILY BE REMEMBERED. LEAVING THE SYSTEM MANAGER ACCOUNT WITH A NULL PASSWORD SHOULD BE AVOIDED. As a protection against unauthorized users gaining access to this account and its privileges, it is suggested that the System Manager password be changed periodically, and that the password not be distributed to other MDSS users.

When using the System Manager account (or any other account with a priority level of 99), some of the functions will operate differently from a normal user account. For example, when initiating a self-test of a control product, the System Manager will see additional information on the display that a normal user would not.

An additional menu item to "Identify current software module" also appears only in the menus of users with a priority of 99. These "extra" features are provided to help GE in diagnosing problems with both the MDSS software and with the process control system itself. This additional information should be reported to GE if problems are ever encountered with the MDSS.

7.2 Setting Up User Accounts And Priorities

The assignment of passwords and priorities is an issue which requires some planning. The MDSS is flexible in that the System Manager can decide not to use any of the protective features built into the software, if desired. The simplest MDSS system would have a null username assigned with no password and a priority of 98. Users could then log into the MDSS by simply pressing the <RETURN> key, and have access to all of the functions in the system. This may be desirable in some applications.

In a more complex MDSS system, the System Manager can assign each individual user of the system a unique username and (optionally) a password, and give each user a priority level according to his capability or expertise with the MDSS and the with the process control system. Each user which has been authorized to use the MDSS in this manner is said to have an "account" on the MDSS. By then adjusting the priorities of the menu functions, the System Manager can control exactly who has access to which functions, and will be provided with a printed log detailing each user's activity on the MDSS. This can be used to trace maintenance procedures and record "who has done what" after a problem has been corrected.

MDSS "user accounts" are established and modified using the System Manager function to "Change username, password, & priority", which is described in a later section of this document. Changing the priority of the items in the menus is accomplished using the function to "Modify menu text & priorities", which will also be described later.

7.3 Configuring An MDSS System

Configuring an MDSS system consists of telling the MDSS what control products are connected to it, and where (on which VAX ports) to find them. Additional information, such as the expected polling response, communications parameters (baud rate, parity, etc.), and polling intervals, must also be defined. It takes alot of time initially to set up the system configuration. This service is usually provided by GE, and the original MDSS software distribution will contain a configuration database for the anticipated system configuration. However, if any new control products are added, or the configuration changes for any reason, the System Manager must modify the configuration database to reflect the changes.

The MDSS system configuration is maintained using the System Manager function to "Modify the system configuration", which is explained in a later section of this document.

7.4 The System Manager Menu

The System Manager Menu is accessed by selecting the "System Manager functions" item in the Main Menu. The System Manager Menu allows the System Manager, and any other users which are authorized to access it, to select from a number of special functions which the normal user does need to access. Each item in the menu is a separate function. Most of the System Manager functions are standard functions and will always be present in the menu, while other functions described in this section may be options which are purchased with the system or added at a later date.

In a MDSS system with a large number of user accounts, the System Manager may want to consider giving other key users access to the System Manager Menu to perform such functions as system backup or modifying the system configuration. When the software is distributed, a priority of 50 is assigned to the System Manager Menu; any user with a priority of 50 or higher will have access to the System Manager Menu, but may not have access to all of the functions described below. This can be changed by the System Manager, if desired.

7.5 Standard System Manager Menu Functions

The functions listed in this section are always part of an MDSS system. These functions are all required to control basic MDSS activities. Depending on how the System Manager assigns priorities to these menu items and to each user, all users may not have access to all System Manager functions. Most users will probably not require access to ANY of these functions, and should not even have access to the System Manager Menu item in the Main Menu.

7.5.1 Changing Username, Password, & Priority

This function is used to set up MDSS user "accounts". An account consists of a user name, optional password, and a priority number between 1 and 99. The MDSS software is initially distributed with a single account - the System Manager user, with a username of "SYSTEM MANAGER", a priority of 99 and a null password. All other accounts must be set up by the System Manager after the software has been installed. One of the first functions performed by the System Manager should be to change the System Manager password. The System Manager can then assign other usernames and passwords using this function.

Selecting this function will place the user in the User Authorization Menu. The following functions are available:

1. Exit - return to System Manager Menu

   This function exits the User Authorization function and returns the user to the System Manager Menu.

2. Add a new user

   This function allows the user to add new user accounts to the MDSS.

3. Delete an existing user

   This function allows the user to delete existing user accounts from the MDSS.

4. Change an existing user

   This function allows the user to modify the password and/or priority value of an existing user account.

5. Display existing user(s)

   This function will display the parameters for a single user account or for all user accounts.

6. Print list of all users

   This function will create a printed list of the username, password, and priority for all MDSS user accounts. The passwords of users with higher priority than the current user will not be displayed.

Once a menu item is selected, the user is prompted for a user name to perform the selected function on. See "How To Use the Prompts" for a complete description of how to enter and edit the user name. If the Display or Print function is selected, ths user will be asked to either enter a username or to press the <RETURN> key to display ALL users. Note that when changing the System Manager account, the user is not prompted for a new priority, since the System Manager account is fixed at a priority of 99. It is also not possible to delete the System Manager account from the system.

User names can be up to 15 characters long, and may contain any combination of letters, numbers, spaces, and special characters. Passwords have the same restrictions, and can also be up to 15 characters long. User priorities can be any numeric integer value between 1 and 99. It is not possible to change a user name (using the Change function) once the account has been established. To do this, the old account must be deleted, and a new user added to the system, using the Add function.

The System Manager has access to ALL user accounts and information. If a user with lower priority (say 75, for example) is given access to this function, that user would only have access to other users with priorities of 75 or lower. In other words, giving a user with a priority of 75 access to this function would NOT allow him to display or modify the System Manager account, or any other accounts at a higher priority. That user would also not

be able to add any new accounts with a priority higher than 75. This allows the System Manager to authorize one or more users with priorities less than 99 to maintain the MDSS user accounts, without having to give total access to the MDSS via the System Manager account.

The MDSS system is designed to handle only one user with a null user name, but any number of users may be assigned a "null" password. A "null" assignment is defined as a blank data field. If a user is assigned a null password, then that user will not be asked to enter a password while logging into the MDSS from the Monitor Mode. If the null username feature is used, it is recommended that the null user be given a low priority, since anyone will be able to gain access to this account by simply pressing the RETURN key. It is possible to have a null user in addition to many other "non-null" users.

7.5.2 Modifying The System Configuration

The MDSS maintains information about every control product connected to it. This information controls such things as communication parameters, polling parameters, and product names. The user should not attempt to modify any parameter without first completely understanding what the parameter means. The most common parameters which a user might want to modify are the product name and product description. The software is distributed with a default product name and description assigned to each product; the user may change these to more meaningful names if desired, using the "Change" function described below.

The entire MDSS configuration is stored in a file called the Configuration Database. When the Configuration Editor is invoked, this database is loaded so that the editor has access to all of the system parameters. Note that the MDSS software, when it is running, makes a copy of this database, and uses the copy for polling and communications. Therefore, the configuration editor can access the original database and make changes without effecting the running system. The new parameters will not take effect until the user selects the "Reset configuration" function in the System Manager Menu (or reboots the VAX).

Selecting the Configuration Editor presents the user with a System Configuration Editor menu containing the following functions:

1. Add a new control product

   This function allows the user to add a new control product to the MDSS configuration database.

2. Change a control product

   This function lets the user modify parameters for a control product which already exists in the configuration database.

3. Delete a control product

   This function will allow the user to remove all parameters for a control product from the database.

4. Display a control product

   This function will display all of the parameters for the selected control product to the video terminal.

5. Print the system configuration

   This function will print a listing of all the control products which are in the configuration database along with some of the critical parameters for each product.

6. Change polling delay for the system

   This function allows the user to change the system polling delay interval for the whole MDSS system.

The polling delay for the system determines the interval at which the MDSS polls the products. For example, if there are ten control products in the system and the polling delay is set to 6 seconds, the MDSS would poll one product every 6 seconds, and each product would be polled once every minute. Note that polling is completely independent of event logging. All events reported to the MDSS are logged as soon as they are received. Polling is a background function which simply asks a product if it is

still connected to the link and able to communicate (see "Polling of Control Products" for more information on polling).

The minimum polling delay between products is 4 seconds. This limit is required to maintain reasonable response time of the MDSS to user input and event messages. The longer the polling delay, the better the response time. The "Change polling delay" function of the Configuration Editor allows the user to alter this parameter; any value between 4 and 99 seconds will be accepted.

The other five menu items (Add, Change, Delete, Display and Print) call the product selection menu which allows the user to select a control product to perform the requested function on. No function can be performed until the user has selected a control product from this menu.

The following parameters are contained in the configuration database for each product:

1. PRODUCT NAME

   This is a 3-6 character alphanumeric name assigned to each product. This name will used to identify the product on the video terminal and printer, and must be unique for each product in the system.

2. PRODUCT TYPE

   This is a 6-character code which identifies the type of control product and what version software it contains. For example, a type of "DMCV02" is a DMC with version 2 software. The first three characters must be one of the valid product types known to the system.

3. PORT ID NUMBER

   This is an integer number denoting which logical name points to the physical VAX port which the control product is connected to. Logical names are of the form MDSS_PORT_nn, where "nn" is the port id number.

4. PORT BAUD RATE

This is the baud rate at which the MDSS will communicate with the control product. Valid baud rates are: 50, 75, 110, 134, 150, 300, 600, 1200, 1800, 2000, 2400, 3600, 4800, 7200, 9600, and 19.2 (19,200). To keep the interrupt load on the VAX within limits, it is strongly recommended that the baud rate not exceed 1200 for any product directly connected to the MDSS.

5. PARITY (E,O,N)

This determines the sense of the parity bit transmitted with each character. Some control products require even parity, some require odd parity, and some do not require any parity. Valid choices are: "E" for Even, "O" for Odd, and "N" for None.

6. DATA BITS (7,8)

This determines the number of data bits per character between the MDSS and the control product. Valid choices are 7 or 8 bits.

7. WAKEUP SEQUENCE

This ascii string defines the character(s) to be sent to the control product to "wake it up" when it is offline. Typically, most autobaud products wake up to a series of RETURN characters (^M); non-auto-baud products usually wake up to a single Control/C (^C). Control characters can be entered into this string by entering the two-character sequence "^C" for Control/C, etc. This sequence will be translated into a single ascii 3 internally.

In addition, a "quit sequence" may be defined by placing a "|" character at the end of the wakeup sequence, then following it with the quit sequence. A quit sequence will be transmitted to the control product one polling delay after it comes online (or is released from direct access) to cause it to go to some predefined state. The "|" character is a separator and is never transmitted.

8. POLLING RESPONSE

This is the ascii string that the product responds with when it is polled. The string will depend on the type of control product. Any message received from the control product which contains the polling response string is considered to be a valid polling response and is NOT logged to the printer.

9. DESCRIPTION

This is a 1-18 character string which describes the control product. It is used in the product selection menus to help the user identify the control products.

10. DOWNLOAD FILENAME

This is the filename of the software file which should be downloaded to the control product by the optional DOWNLOAD function. You should normally leave this field blank, unless you intend to download this particluar control product using the MDSS. If you define a filename for the product, then the product will appear in the menu of control products for the download function. If you leave this field blank, the control product will NOT appear in the menu.

11. ON-LINE POLLING RATIO

This parameter determines how often in the polling cycle the product gets polled when it is on-line. If the ratio is 1 (the usual value), the product is polled each time through the cycle. If the polling ratio was 3, the product would only be polled every third time around the polling cycle.

12. OFF-LINE POLLING RATIO

This parameter is the same as the on-line polling ratio, except it determines how ofter a product is polled when it is off-line. Usually when a product is off-line, it should be polled less often than those products still on-line. A typical value of this parameter is 3.

13. MAXIMUM ERRORS ALLOWED

This determines how many times a product can not respond (or respond incorrectly) to a polling request before an event message is logged and the product is taken offline. A typical value is 2 or 3; a value of 1 could cause false event messages, since it is possible for a product to miss a polling response occasionally due to higher priority control activity, or noise on the MDSS link.

14. WAKEUP SEQUENCE DELAY

This value is the delay between wakeup sequence and quit sequence characters, specified in milliseconds. For non-auto-baud products with no quit sequence, this value should be 0; auto-baud products have a typical value of 10-50 milliseconds.

Some control products have an "auto-baud" feature. This means that after applying power to these control products, they will look for a particular character at many baud rates, and automatically lock onto the correct baud rate when the specified input character is received. This feature requires special logic in the MDSS to force the control product to communicate with the MDSS after a powerup. The "wakeup sequence" parameters listed above are used by this special logic to "wake up" the control product after a powerup or other off-line condition by sending (typically) a series of carriage return ascii codes (^M^M^M...) to the product. The delay between each ^M code is defined by the "wakeup sequence delay" parameter.

For non-auto-baud control products, the "wakeup" sequence is simply a polling request (a single ^C code). A quit sequence is typically used to exit the monitor of a control product such as the DMC. A typical quit sequence for the DMC would be a "Q" followed by a carriage return ("Q^M") to quit the monitor.

When the Add or Change functions are selected, current parameters for the control product are displayed. (In the case of Add, default parameters are displayed). The user can move the cursor from parameter to parameter using the <up arrow> and <down arrow> keys, leaving the parameters unchanged. Any parameter can be modified by simply moving the cursor to that field and changing the parameter value.

To delete the old parameter value and enter a new value, press <CTRL/D> when the cursor is on the correct field. After the new value for a parameter has been entered, the user must press the <RETURN> key to actually make the change. See the section "How To Use the Prompts" for more information.

When the desired parameters in the list have been modified, the user can return to the Configuration Editor Menu by either moving the cursor past the end of the list (using the <RETURN> or <down arrow> key), or by pressing <CTRL/A> after the last parameter has been entered to abort changing the rest of the parameters in the list.

While modifying the configuration parameters, no changes are saved in the database until the user exits the program. Upon exiting, if any changes have been made to the data, the user is asked if the changes should be saved. Answer <Y> to update the database with all changes made, or <N> to leave the database unchanged.

7.5.3 Backing Up The MDSS Software

This function allows backup of the MDSS data files. The MDSS data files can be backed up onto any removable media, or the entire MDSS system disk can be backed up. Also, in the event that an MDSS data file should become lost or corrupted, a function to restore the MDSS data files is also provided.

Selecting this function will place the user in the MDSS Disk Backup Menu. The following functions are available:

1. Exit to the System Manager Menu

   This function will exit the backup menu and return the user to the System Manager Menu.

2. Backup message tables and data files

   This function will copy only the MDSS data files to the backup media.

3. Backup the entire system disk

   This function will make an image backup of the entire MDSS system disk. Note that this function can take a considerable amount of time to execute (a minimum of 20 minutes should be allowed). Since the system will be active during this function, several warning messages may be displayed by the backup function about files being in use - ignore these messages. They will not affect the backup copy.

4. Restore message tables & data files

   This function will copy the MDSS data files from a previously-written backup copy to the MDSS system disk.

Whenever a backup or restore operation is completed, a message is logged on the printer with the date and time, type of operation performed, and the user name of who initiated the operation. It is not possible to perform a complete disk restore using the MDSS backup function, since the MDSS software would have to be resident on the disk and active to do this. You must use the VAX BACKUP utility to restore the entire system disk.

Note that the device which is used by the MDSS software to make backup copies of the disk is defined by the logical name MDSS_TAPE_1 in the file CONFIGURE.DAT located in the directory MDSS$_Sysdir:.

For additional information on backing up the MDSS system, please refer to the section on "Backing Up the VAX System Disk" in this document.

7.5.4 Stop Polling Of Control Products

This function suspends the polling of all of the control products connected to the MDSS. When the process control system is not operating and many of the control products may be powered down for maintenance or other reasons, this function can be used to suppress the "Control product ONLINE/OFFLINE" messages which would be generated. It also eliminates all of the MDSS background activity on the VAX, if this should become desirable. Note that unsolicited event messages which originate in the control products will still be logged even after polling has been

suspended.

Selecting this function will cause the MDSS to ask the user "Are you sure you want to stop polling?". The user should answer <Y> or <N>. A <Y> response will suspend the polling; either response will return the user to the System Manager Menu.

7.5.5 Reset Configuration And Start Polling

This function restarts the polling sequence on all current control products known to the MDSS. This function can be used after the polling has been suspended (via the "Stop Polling..." function), or after the system configuration has been modified (via the System Configuration Editor).

When the MDSS software is started after rebooting the VAX, a copy is made of the system configuration table, and the MDSS system uses this copy for all communications. If the configuration editor is invoked to modify the table, the original table is modified, NOT the copy that the system is currently using. Once the original table has been modified, this "Reset" function can be invoked to make a new copy of the original table for the MDSS to use for its communications. Selecting this function, then, is the equivalent of "rebooting" the MDSS software.

7.5.6 Adding A New MDSS Software Module

This function allows MDSS users to add new MDSS modules to their system as they become available. MDSS has been designed as an expandable system which can grow as new modules (functions) are defined and developed. In fact, it is possible for MDSS users to create their own function modules and add them to the system; this is another option available for MDSS.

When a new MDSS module is purchased, or GE sends an update distribution for the current MDSS software, this function will be used to install the software. You will be supplied with some distribution media (usually a magnetic tape) and a "module name". When this function is selected, the MDSS will prompt for the module name - enter the 6-character module name which was supplied with the distribution media. Then simply follow the instructions on the terminal; load the media on the correct device, press the <RETURN> key, and wait a few minutes until the function completes and returns to the System Manager Menu.

The new module is now installed.

When the function completes, a message will be logged to the printer giving the module name and the name of the user who installed it. If the media contained a new MDSS module, it will automatically appear as a new item in the correct menu. It is usually not necessary to reboot the VAX or even restart the MDSS software to complete the installation, unless specifically instructed to do so.

Note that the device which is used by the MDSS software to add new software modules is defined by the logical name MDSS_TAPE_1 in the file CONFIGURE.DAT located in the directory MDSS$_Sysdir:.

7.5.7 Modifying Menu Text & Priority

This function can be used to change the priority of any item in the MDSS Main Menu or the System Manager Menu. It can also change the wording of the items in those menus, if the original wording is not satisfactory. If the user creates a new MDSS module, this function can also be used to install the new module into one of the MDSS menus. (This requires the MDSS user-created module option).

Selecting this item causes the MDSS to ask "Do you want to change Main Menu entries (Y,N) ?" - answer <Y> if you want to change items in the Main Menu, or <N> to change items in the System Manager Menu.

Once a menu has been selected, the first item for that menu is displayed, and the user is asked "Do you want to change this entry (Y,N) ?". Answer <Y> to change either the text or the priority, or <N> to go on to the next menu item.

Only menu items which currently have a priority less than or equal to the user's own priority will be displayed.

If a <Y> response is given, the text for the menu will be displayed and the user may modify it if desired, using the prompting function. Press the <RETURN> key when the text is correct. Then the user will be prompted for the new priority of the menu item. Enter a new priority value, if desired, by first deleting the old value (press <CTRL/D>), then typing a new value, again using the prompting

function.

When the priority is correct, again press the <RETURN> key to go on to the next menu item. At any point, the user can press <CTRL/A> to skip to the end of the list of menu items.

After all items are reviewed, new items may be added to the menu. (This is used by the "Add a new module" function to add entries to the menu). The user should answer <N> to the question about adding additional items, unless the user-created module option has been purchased and the user is adding a new module to the system. Upon completion of all changes the user is prompted for a decision on whether the changes made are to be permanently saved or not. Answer <Y> to save the changes to the menu file, unless you do not wish to save the changes made.

If the changes are saved, they will immediately become effective. For example, if the System Manager Menu was being modified, when the function exits and returns to the System Manager Menu, any changes which were made should be present at that time, and will affect all future users of the system.

7.6 Optional System Manager Menu Functions

Currently all System Manager functions are provided with the basic MDSS software. In the future, new functions may be developed which could be added to existing MDSS systems as options (using the "Add a module" function described above).

INDEX

aborting functions, 5, 13, 15, 18, 27
adding new functions, 67
alarm annunciation, 8, 13, 15, 24, 31 to 32
alarm delay time, 32
application code, 22, 41
application message, 20, 22
application message editor, 40
arrow keys, 3, 5 to 6, 29, 39, 64
auto-baud, 64
AUTOGEN, 50
automatic MDSS startup, 51

backup, 51
backup function, 65
beep, 5 to 6, 8, 16, 23

changing menu priorities, 68
circular list plot, 44
configuration database, 59, 61
configuring the system, 51, 56, 59
control product
  definition of, 2
  diagnostics, 4, 22, 35, 43
  identification, 20
  messages, 20
  port, 2, 4
CTRL key, 11, 13, 15
CTRL/A, 5, 13, 15, 27 to 28, 30, 44, 69
  definition of, 1, 18
CTRL/P, 29
cursor, 3, 12, 14
  definition of, 5
  movement of, 5

date & time
  changing, 29
  message stamp, 20
date & time stamp, 7
DELETE key, 1, 11, 14 to 15
DETACHSET file, 50
direct access function, 28
display status function, 43

event messages, 7, 19

functions
  access limitations, 10
  accessing, 10
  selecting, 3, 12, 26

keyboard, 5
  (see also video terminal)
  basic keys used, 6, 14
  YES/NO response, 6

link, 3 to 4
local terminal, 3 to 4, 28
log-in, 8 to 11, 54
log-off, 27
logging in, 7

MDSS
  adding new functions, 4, 67
  configuration database, 59
  definition of, 2
  link, 3
  loading software, 49
  menus, 3, 26
  overview, 1 to 2
  starting software, 7
  stoping software, 9
  System Manager, 1, 10
  testing software, 51
MDSS internal message, 11
menu-driven, 3
menus
  format of, 12
  selecting items in, 12 to 13
  use of, 3, 12, 26
message explain editor, 37
message explain function, 36
message format
  date stamp, 20
  line of dashes, 24
  message code, 20, 41
  message text, 21, 42
  product identification, 20
  time stamp, 20 to 21
message log files, 32, 34

messages
  application, 20, 22
  event, 4, 22
  explaining, 36
  logging, 27, 66, 68
  MDSS internal, 11, 22 to 23
  message code, 36
  startup, 7
micro-processor, 2
Monitor Mode, 7
  description, 8
  event message display, 8, 23, 27
  message scrolling, 8
  returning to, 27
  timeout, 10
multi-users, 9, 29, 31

null
  password, 59
  username, 10, 55, 59

offline/online, 19, 35, 43, 66

password, 11, 17, 58
PF4 key, 8, 13, 15, 31 to 32
plot circular list, 44
polling, 19, 60
  starting, 67
  stopping, 66
polling delay, 61
PRINT SCREEN key, 25, 45
printer, 7, 11, 19, 24
printer port, 25
priorities
  changing menu items, 68
  how they work, 17
  range, 17
privileges
  (see priorities)
process control system, 2
prompt, 10 to 11
prompt function, 4, 12, 14, 65
protection, 3

RETURN key, 1, 5 to 6, 11 to 13, 15
reviewing log files, 32, 34
RS-232, 4

selftest function, 35
SETUPMDSS file, 50
software
  loading, 49
  starting, 7
  stoping, 9
  testing, 51
SYSTARTUP.COM file, 50
System Manager
  access, 58
  account, 54
  function of, 1, 10, 54
  menu, 29, 56 to 57
  password, 54
  priority, 17, 45, 54, 57
  user, 54

timeout, 10

unsolicited event messages, 4
user
  accounts, 55, 57
  password, 17
  priority, 17
  System Manager, 10
  unauthorized, 54
user authorization function, 57
username, 10, 58

VAX computer, 1
  backing up the disk, 51, 66
  login, 7, 49, 51
  minimum configuration, 48
  rebooting, 7, 29, 67 to 68
video terminal
  arrow keys, 5 to 6, 39, 64
  beep sound, 5 to 6, 8, 16, 23
  CTRL key, 1, 5, 11, 13, 15, 18
  cursor, 5, 12, 14
  DELETE key, 1, 11, 14 to 15
  keyboard, 5
  Monitor Mode, 8
  RETURN key, 1, 5 to 6, 11 to 13, 15
VMS operating system, 1
VT220 terminal
  (see video terminal)

YES/NO questions, 6